

Algorithm	RRT				RRT*			
Test environment	Execution time median [s]	Path length median	Vertices count median	Reliability	Execution time median [s]	Path length median	Vertices count median	Reliability
A	137.0	22	569	75%	95.6	19	107	90%
B	134.1	20	538	85%	116.5	21	109	85%
C	159.6	22	658	75%	72.9	19	110	90%
Median	137.013	22	569	75%	95.5765	19	109	90%

Detailed results 1		Environment A					
		RRT			RRT*		
Run		Time	Path length	Vertices count	Time	Path length	Vertices count
1		207.015	23	740	58.315	17	101
2	X	300.067	X 24	X 1815	65.383	18	121
3		108.538	22	296	140.563	19	149
4		137.013	21	971	156.55	21	121
5		80.016	22	295	140.212	18	176
6	X	300.074	X 23	X 2349	64.48	19	106
7		182.003	20	227	47.535	17	107
8		166.067	23	339	101.567	19	86
9		94.063	20	197	73.456	19	101
10		139.124	22	842	219.974	19	80
11	X	300.131	X	X	86.179	22	83
12		225.661	22	517	89.586	23	227
13		105.373	25	569	122.579	19	76
14		112.949	24	625	55.457	21	116
15		84.626	21	272	127.031	22	195
16	X	300.075	X	X	114.855	24	96
17		133.39	20	962	X 301.221	X	X
18		300.074	20	2560	X 301.233	X	X
19		232.795	19	815	60.92	20	86
20	X	300.045	X 23	X 1351	159.004	32	213
Median		137.013	22	569	95.5765	19	107
Valid runs		15			18		

X – terminated due to timeout and/or failure

Environment B					
RRT			RRT*		
Time	Path length	Vertices count	Time	Path length	Vertices count
61.37	20	140	151.888	21	203
194.547	16	258	49.792	20	53
X 300.048	X 27	X 1756	127.08	20	57
80.268	24	442	X 301.255	X	X
X 300.072	X 22	X 1905	78.865	22	217
135.502	24	610	195.244	22	80
144.775	22	409	X 301.307	X	X
114.9	20	836	41.675	18	95
141.42	20	362	168.941	28	224
219.919	27	769	55.26	22	203
124.633	20	976	115.952	20	140
225.78	24	671	116.502	21	272
120.113	21	156	97.158	17	59
235.205	24	1014	70.896	23	161
X 300.109	X 25	X 2630	X 301.264	X	X
134.097	20	469	133.594	22	109
84.629	20	538	157	18	113
221.761	20	804	118.762	18	47
125.029	23	190	90.192	24	76
106.04	19	853	213.641	20	93
134.097	20	538	116.502	21	109
17			17		

Environment C					
RRT			RRT*		
Time	Path length	Vertices count	Time	Path length	Vertices count
140.169	21	924	57.962	18	37
67.485	15	296	147.384	16	173
X 300.131	X	X	111.792	18	85
159.563	20	652	56.962	17	91
277.96	27	1296	X 301.196	X	X
95.508	20	534	70.789	19	225
186.52	23	250	126.58	19	98
X 300.229	X	X	101.63	24	116
127.239	22	858	170.298	24	280
227.138	24	857	70.049	20	103
X 300.069	X 19	X 1964	65.681	22	360
276.309	18	1131	X 301.239	X	X
94.804	25	658	55.658	19	224
229.981	24	485	123.422	15	94
77.259	22	277	69.792	19	82
X 300.048	X	X	143.012	25	212
X 300.073	X 23	X 2247	75.056	19	166
213.972	18	1489	63.328	16	102
72.448	24	385	69.188	18	97
243.624	24	784	162.625	26	508
159.563	22	658	72.9225	19	110
15			18		